

Anomaly Detection using Convolutional Autoencoders on Tiny-ImageNet

1 Problem definition & methodology

The objective of this assignment was to investigate whether Convolutional Autoencoders (CAEs) can be used for anomaly detection on Tiny-ImageNet (10 classes).

The task was to:

- Train an autoencoder on 9 classes (seen data)
- Hold out 1 class as OOD (unseen)
- Compare reconstruction error (MSE) between seen and unseen samples

Methodology:

1. Selected one class (e.g., class 3) as OOD
2. Trained a convolutional autoencoder on remaining 9 classes
3. Evaluated reconstruction error on:
 - Seen classes
 - Unseen (OOD) class
4. Visualized:
 - Training loss
 - Reconstruction error per class
 - Original vs reconstructed images
5. Extended experiment by:
 - Repeating training for each class as OOD (10 models)
 - Computing mean and variance of reconstruction error per class

2 Hyperparameter tuning & architecture designs

Model: Convolutional Autoencoder

- Encoder: 3 Conv layers (stride=2), channels: 3 -> 32 -> 64 -> 128
- Decoder: 3 ConvTranspose layers (mirroring encoder)
- Activation: ReLU (hidden), Tanh (output)
- No BatchNorm or Dropout

Training setup:

- Loss: Mean Squared Error (MSE)
- Optimizer: Adam ($lr = 10^{-3}$)
- Batch size: 128
- Epochs: 30

Observation: Training loss converged smoothly, indicating stable learning.

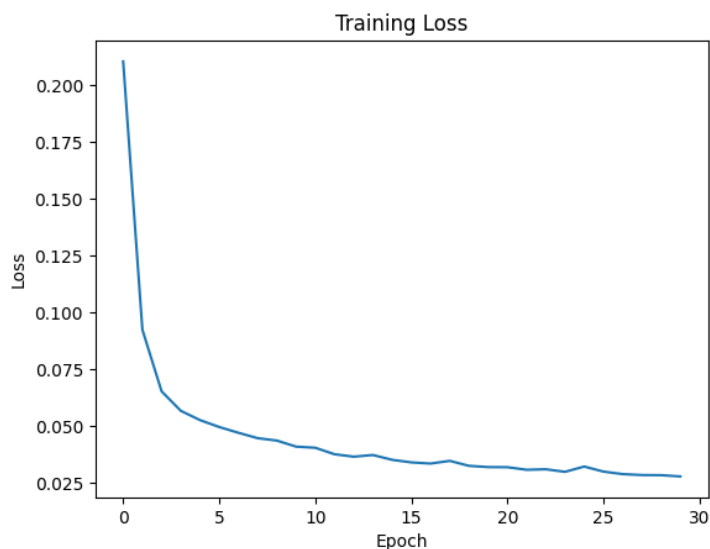


Figure 1: Training loss curve

3 Results, visualizations, and interpretations

Single OOD experiment (Class 3):

- Seen class MSE: 0.0284
- Unseen class MSE: 0.0168

Interestingly, the unseen class did not always produce higher error, indicating that reconstruction alone may not reliably detect anomalies.

Class-wise reconstruction error:

Class 0: Avg MSE = 0.028132
 Class 1: Avg MSE = 0.036198
 Class 2: Avg MSE = 0.018450
 Class 3: Avg MSE = 0.016844
 Class 4: Avg MSE = 0.028276
 Class 5: Avg MSE = 0.027768
 Class 6: Avg MSE = 0.043647
 Class 7: Avg MSE = 0.023515
 Class 8: Avg MSE = 0.020019
 Class 9: Avg MSE = 0.029609

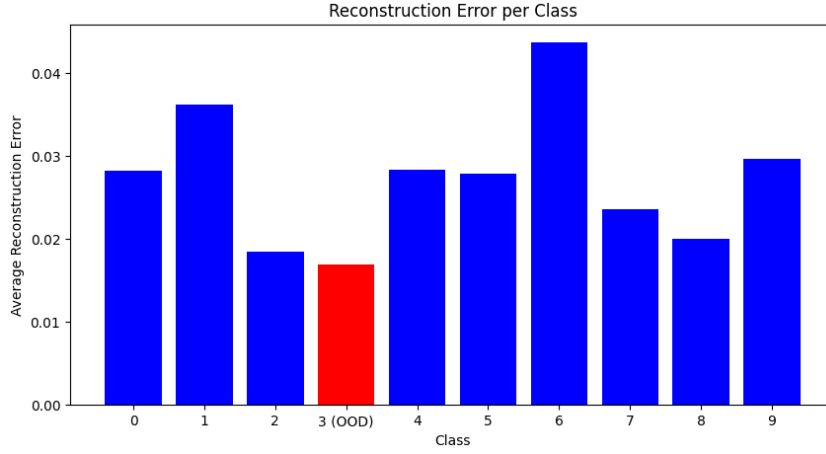


Figure 2: Reconstruction error per class (OOD highlighted)

Reconstruction visualization:



Figure 3: Original vs reconstructed images

The reconstructions are generally good, even for the unseen class, suggesting the model learns shared visual features rather than class-specific patterns.

4 The key findings and explanations

- Autoencoders do not reliably detect anomalies: Reconstruction error for unseen class is not consistently higher.

- Error depends on feature similarity: Classes sharing visual patterns (texture, color, structure) are reconstructed well even if unseen.
- Low variance across OOD runs:

Class 0: Mean=0.043255, Var=0.000002
 Class 1: Mean=0.054327, Var=0.000002
 Class 2: Mean=0.030177, Var=0.000001
 Class 3: Mean=0.028947, Var=0.000001
 Class 4: Mean=0.039932, Var=0.000001
 Class 5: Mean=0.038491, Var=0.000001
 Class 6: Mean=0.063730, Var=0.000005
 Class 7: Mean=0.035143, Var=0.000001
 Class 8: Mean=0.029017, Var=0.000001
 Class 9: Mean=0.048054, Var=0.000008

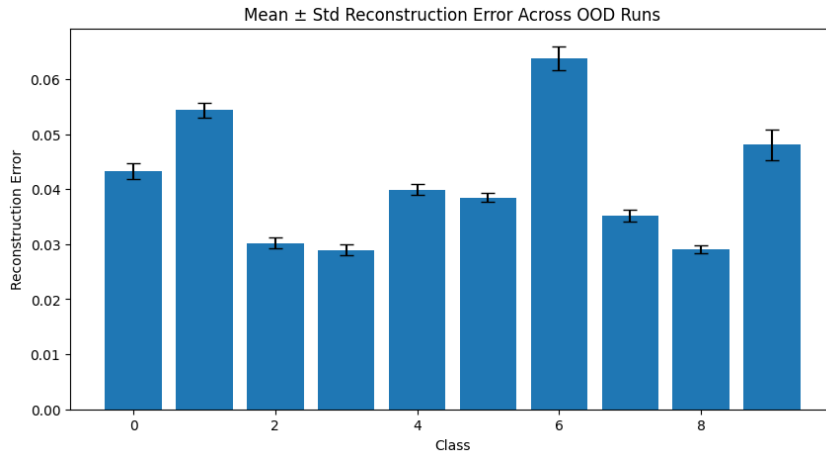


Figure 4: Mean $\pm$ Std reconstruction error across OOD runs

Conclusion:

- Reconstruction error is not strongly dependent on which class is OOD
- CAEs learn a shared latent representation of visual features
- This limits their effectiveness for anomaly detection in complex datasets

Final insight: Anomaly detection using autoencoders works only when anomalies are sufficiently different in feature space, not merely unseen during training.